



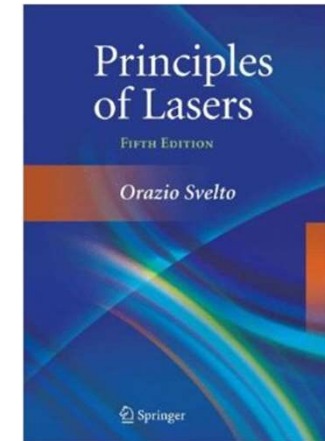
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch5-2

Characteristics of Laser Amplification



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Chapter 5: Amplification Characteristics

❖ 5.3 Gain Characteristics of the Longitudinal Excited Continuous Laser Amplifier

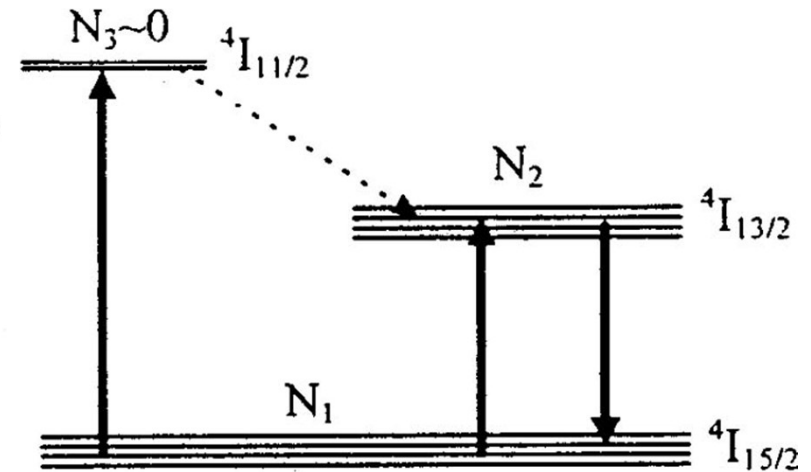
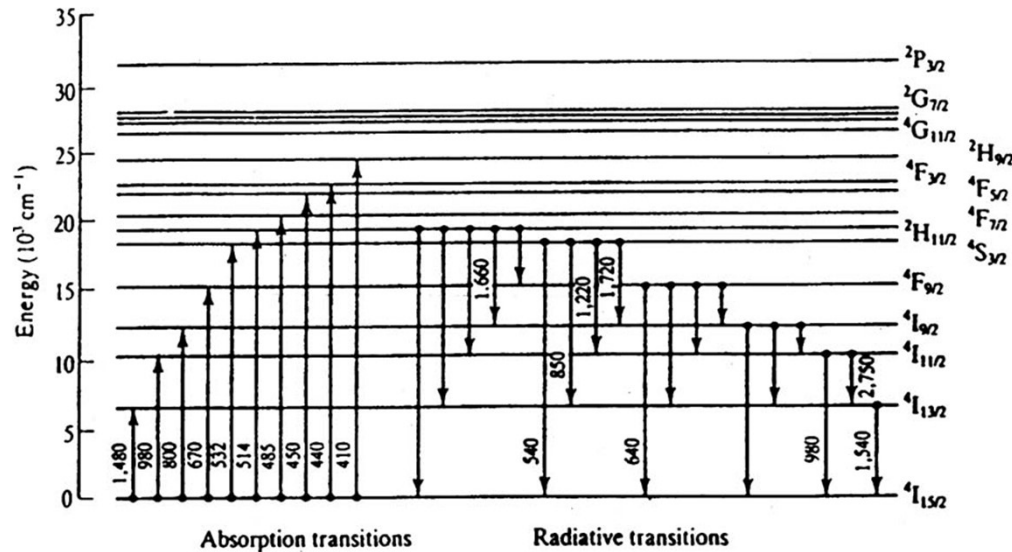
- Signal, pump transport equation
- Weak signal gain characteristics of fiber amplifier
- Intensive signal gain characteristics

❖ 5.4 Gain Characteristics of Pulse Laser Amplifier

- Transport equation of pulse laser amplifier
- Output energy and energy gain of pulse laser amplifier
- Power gain and pulse width narrowing

5.3 Gain Characteristics of the Longitudinal Excited Continuous Laser Amplifier

Erbium-Doped Fiber Amplifier: EDFA



- Why EDFA belongs to continuous laser amplifier?
- ✓ Optical signal transition rate: $10^8 \sim 10^{10} \text{ b/s}$ ($T = 10^{-8} \sim 10^{-10} \text{ s}$), erbium ions: $T_1 = 10^{-2} \text{ s}$, $T \ll T_1$ very short action time, Δn only has small fluctuation near the steady-state value
- Characteristics of longitudinal optical excitation

$$I(z), I_p(z) \Rightarrow g^0(z), \Delta n^0(z), I_s(z)$$

Assume: three-level system, ignore the transverse distribution of the optical field and the activated ions; ignore the fiber loss; the quantum efficiency $\eta_F=1$; homogeneous broadening; pumping in the same direction with signal.

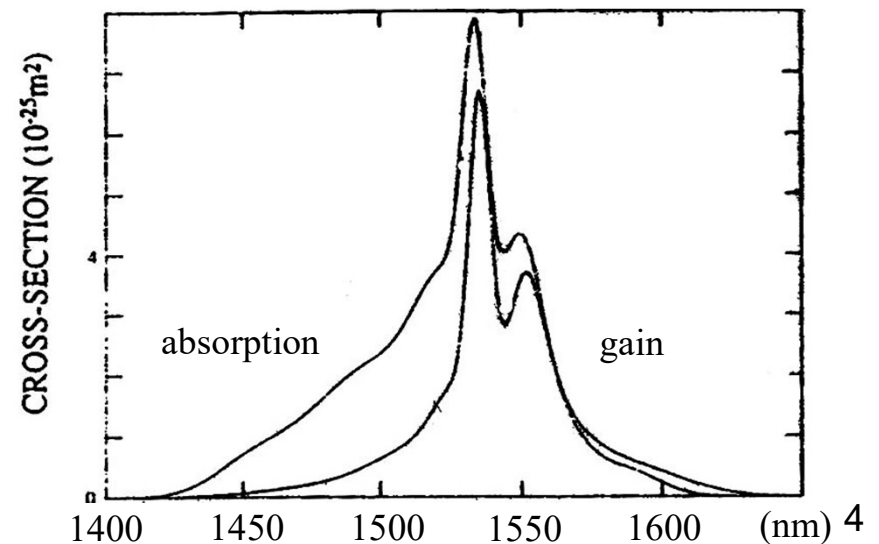
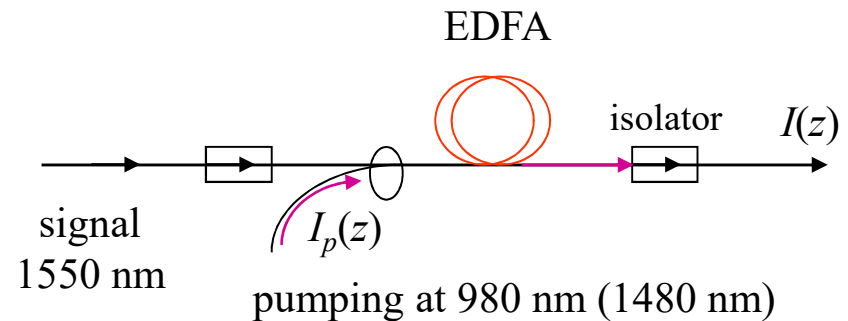
❖ Signal, pump transport equation (ignore SP)

$$\frac{dI(z)}{dz} = \Delta n(z) \sigma_{21}(\nu) I(z)$$

$$\frac{dI_p(z)}{dz} = - \left[n_1(z) \sigma_{13}(\nu_p) - n_3(z) \sigma_{31}(\nu_p) \right] I_p(z)$$

$$\Delta n(z) = n_2(z) - \gamma n_1(z)$$

$$\gamma = \frac{\sigma_{12}(\nu)}{\sigma_{21}(\nu)} = \frac{f_2}{f_1}$$



EDFA steady-state rate equation

$$\left\{ \begin{array}{l} \frac{dn_3(z)}{dt} = \left[n_1(z) \sigma_{13}(\nu_p) - n_3(z) \sigma_{31}(\nu_p) \right] \frac{I_p(z)}{h\nu_p} - n_3(z) S_{32} = 0 \\ \frac{dn_2(z)}{dt} = n_3(z) S_{32} - \Delta n(z) \sigma_{21}(\nu) \frac{I(z)}{h\nu} - \frac{n_2(z)}{\tau_s} = 0 \\ n_1(z) + n_2(z) + n_3(z) = n \end{array} \right. \quad \begin{array}{l} I(z) = N\nu h\nu \\ S_{32} \gg 1 \Rightarrow n_3 \approx 0 \end{array}$$

$$\Delta n(z) = \frac{I'_p(z) - \gamma}{I'_p(z) + (1 + \gamma)I'(z) + 1} n \quad n_1(z) = \frac{I'(z) + 1}{I'_p(z) + (1 + \gamma)I'(z) + 1} n$$

Normalized signal intensity $I'(z) = I(z) / [h\nu / \sigma_{21}(\nu) \tau_s]$

Normalized pump intensity $I'_p(z) = I_p(z) / [h\nu_p / \sigma_{13}(\nu_p) \tau_s]$

$$\boxed{\begin{array}{l} \frac{dI(z)}{dz} = \Delta n(z) \sigma_{21}(\nu) I(z) \\ \frac{dI_p(z)}{dz} = - \left[n_1(z) \sigma_{13}(\nu_p) - n_3(z) \sigma_{31}(\nu_p) \right] I_p(z) \end{array}} \Rightarrow \boxed{\begin{array}{l} \frac{dI'(z)}{dz} = \dots \\ \frac{dI'_p(z)}{dz} = \dots \end{array}}$$

Normalized signal, pump light transport equation

$$I'_p(z) = \frac{I_p(z)}{h\nu_p / \sigma_{13}(\nu_p) \tau_s} = \gamma \frac{I_p(z)}{I_{pth}}$$

$$\frac{dI'(z)}{dz} = \frac{I'_p(z) - \gamma}{I'_p(z) + (1 + \gamma)I'(z) + 1} \frac{\beta}{\gamma} I'(z) \quad (5.3.11) \quad \text{Threshold pumping intensity}$$

$$\frac{dI'_p(z)}{dz} = - \frac{I'(z) + 1}{I'_p(z) + (1 + \gamma)I'(z) + 1} \beta_p I'_p(z) \quad (5.3.12)$$

$$I_{pth} = \gamma \frac{h\nu_p}{\sigma_{13}(\nu_p) \tau_s}$$

$$\left(\frac{I_p(z)}{I_{pth}} - 1 \right) \gamma \begin{cases} I_p(z) > I_{pth} \Rightarrow \frac{dI'(z)}{dz} > 0 & \text{Optical signal amplification} \\ I_p(z) < I_{pth} \Rightarrow \frac{dI'(z)}{dz} < 0 & \text{Optical signal attenuation} \end{cases}$$

$$\frac{dI'_p(z)}{dz} < 0$$

pumping light along the fiber ↓

$$\left. \begin{aligned} \beta &= n\sigma_{12}(\nu) && \text{Weak signal absorption coefficient of signal light in doped fiber} \\ \beta_p &= n\sigma_{13}(\nu_p) && \text{Weak signal absorption coefficient of pump light in doped fiber} \end{aligned} \right\} \begin{array}{l} \text{can be measured} \\ \text{experimentally} \end{array}$$

The threshold pump intensity is an indicator of the increase or attenuation of the signal light in the optical fiber.

❖ Weak signal gain characteristics [$I'(z) \ll 1$]

Eq. 5.3.12 $\xrightarrow[\text{integral}]{I'(z) \ll 1}$ Normalized the output pump intensity

$$\ln \frac{I'_p(l)}{I'_{p0}} + [I'_p(l) - I'_{p0}] = -\beta_p l$$

$\frac{\text{Eq. 5.3.11}}{\text{Eq. 5.3.12}} \xrightarrow[\text{Eq. 5.3.12}]{\text{integral}}$ Normalized the output pump intensity

$$\gamma \frac{\beta_p}{\beta} \ln \frac{I'(l)}{I'_0} = \gamma \ln \frac{I'_p(l)}{I'_{p0}} + I'_{p0} - I'_p(l)$$

$$\left. \begin{aligned} G^0^{-\frac{\gamma \beta_p}{\gamma+1 \beta}} &= \left(\frac{I'_{p0}}{I'_p(l)} \right) \exp \left(-\frac{\gamma}{\gamma+1} \beta_p l \right) \\ I'_p(l) &= I'_{p0} - \frac{\gamma}{\gamma+1} \beta_p l - \frac{\gamma}{\gamma+1} \frac{\beta_p}{\beta} \ln G^0 \end{aligned} \right\}$$

multiply the two equations

Can be experimentally measured

$$\ln \left[\gamma \frac{I'_{p0}}{I_{pth}} - \frac{\gamma}{\gamma+1} \beta_p l - \frac{\gamma}{\gamma+1} \frac{\beta_p}{\beta} \ln G^0 \right] + \frac{\gamma}{\gamma+1} \beta_p l - \frac{\gamma}{\gamma+1} \frac{\beta_p}{\beta} \ln G^0 = \ln \left(\gamma \frac{I'_{p0}}{I_{pth}} \right)$$

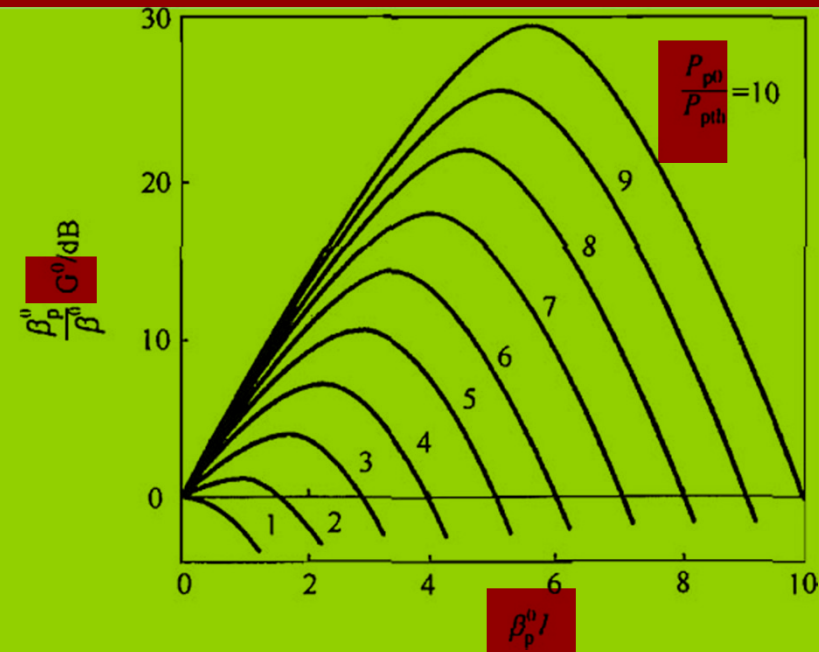
Can be experimentally measured

G^0 is independent of signal intensity, it is related to the pump intensity I_{p0} and the length l of the Er-doped fiber.

Optimal length l_m

- If the length of the fiber is too long, $I_p < I_{pth}$, the signal will decrease
- If the length of the fiber is too short, I_p not fully utilized

when $l = l_m, I_p(l_m) = I_{pth} \Rightarrow I'_p(l_m) = 1$



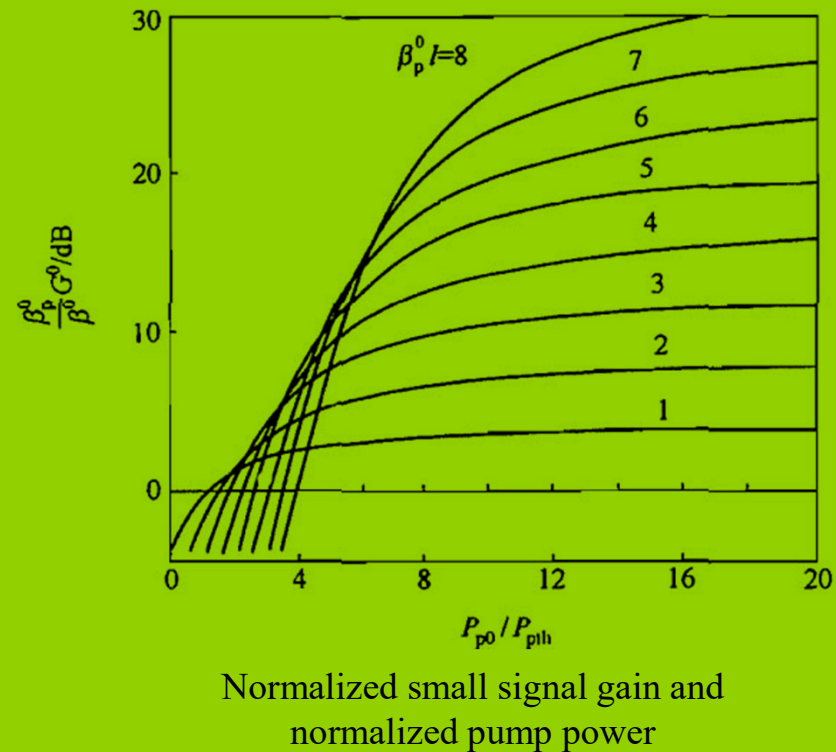
Normalized small signal gain and normalized Er-doped fibre length

$$\frac{I_{p0}}{I_{pth}} - \frac{1}{\gamma + 1} \beta_p l_m - \frac{1}{\gamma + 1} \frac{\beta_p}{\beta} \ln G_m^0 = 1 \quad \Rightarrow \quad l_m = \frac{1}{\beta_p} \left[\ln \frac{I_{p0}}{I_{pth}} + \gamma \left(\frac{I_{p0}}{I_{pth}} - 1 \right) \right]$$

weak signal gain
under optimal length

$$\ln G_m^0 = \frac{\beta}{\beta_p} \left[\frac{I_{p0}}{I_{pth}} - \ln \frac{I_{p0}}{I_{pth}} - 1 \right]$$

- At the very beginning (for weak signal), the gain increase with the optical pumping.
- With increase of pumping power (for strong signal), flattening of the curves have been observed.
- The flattening of the curve implies that the pumping light escapes from the end of the fiber without being absorbed.



Conclusion

- Existence of optimal length, input pump power $I_{p0} \uparrow \rightarrow l_0 \uparrow, G_m^0 \uparrow$
- For fiber with fixed length, if the pumping is too strong, excess pump light will escape from the output end. Then there will be no contribution to the gain coefficient.

❖ Intensive signal gain characteristics [$I'(z)$ can not be ignored]

Characterization of EDFA high power output capability

$$\begin{aligned}\Delta n(z) &= \frac{I'_p(z) - \gamma}{I'_p(z) + (1 + \gamma)I'(z) + 1} n \\ &= \frac{[I'_p(z) - \gamma] / [I'_p(z) + 1]}{1 + \frac{(1 + \gamma)I'(z)}{[I'_p(z) + 1]}} n\end{aligned}$$

$$\Delta n^0(z) = \frac{I'_p(z) - \gamma}{I'_p(z) + 1} n$$

$$I'(z) = I(z) / [h\nu / \sigma_{21}(\nu) \tau_s]$$

$$I'_p(z) = I_p(z) / [h\nu_p / \sigma_{13}(\nu_p) \tau_s]$$

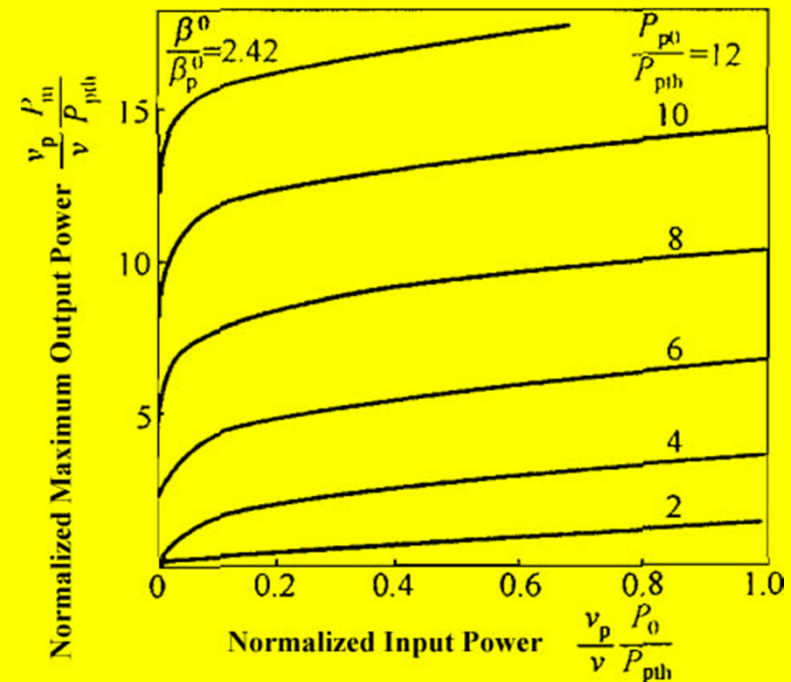
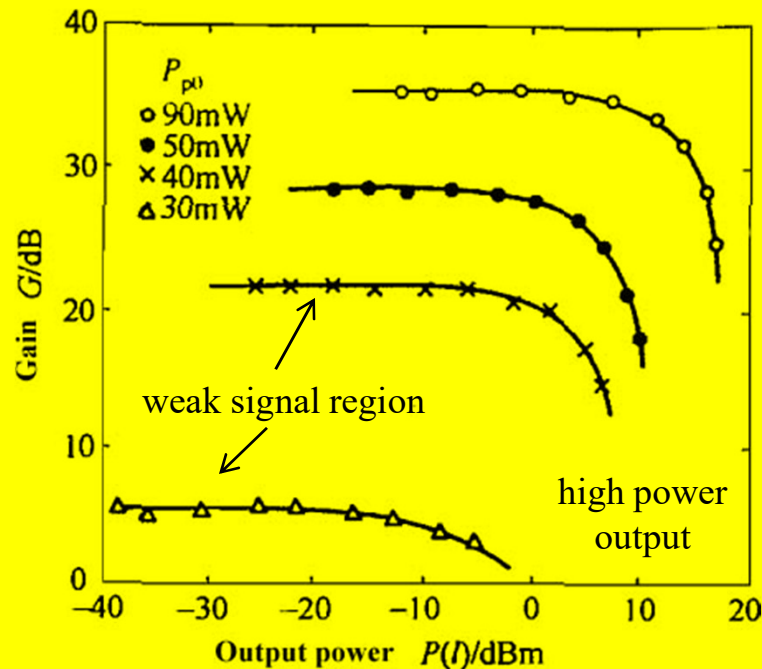
$$\Delta n(z) = \frac{\Delta n^0(z)}{1 + \frac{I(z)}{I_s(\nu, z)}} = \frac{\Delta n^0(z)}{1 + \frac{1}{1 + \gamma} \left[\frac{\nu}{\nu_p} \frac{\sigma_{13}(\nu_p)}{\sigma_{21}(\nu)} \frac{I_p(z)}{I_p(z) + \frac{h\nu}{\sigma_{21}(\nu) \tau_s}} \right]} = I_s(\nu, z)$$

- Saturation intensity is not a constant, it depends on pumping intensity.
- Longitudinal pumping characteristics: $I_p(z)$ change with propagate distance, $I_s(\nu, z)$, $g^0(z)$ also change with distance, $I_p(z)$ relates to the signal intensity $I(z)$.

Numerical solutions are usually required, to get large signal G and output power from transport equation

- At maximum pumping power, calculate the gain and output power at $l=l_m'$ (optimal length under strong signal). Use the same method as the discussion of weak signal case.

$$\frac{\nu_p}{\nu} (G_m - 1) I_0 + I_{pth} \left[\frac{\beta_p}{\beta} \ln G_m + \ln \frac{I_{p0}}{I_{pth}} \right] = I_{p0} - I_{pth}$$

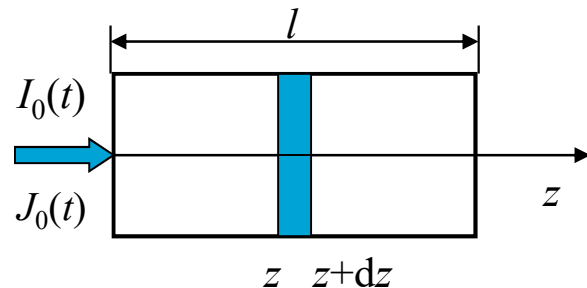


5.4 Gain Characteristics of Pulse Laser Amplifier

Simplified assumption

- Neglecting the influence of optical pumping and SP on population inversion
- Gain material with homogeneous broadening and the incident light $\lambda = \lambda_0$
- Δn has uniform transverse distribution

❖ Transport equation of pulse laser amplifier $\tau_0 < T_1$



$$\frac{\partial N(z, t)}{\partial t} S dz dt = [N(z, t) - N(z + dz, t)] S v dt + \sigma_{21} \nu \Delta n(z, t) N(z, t) S dz dt - \alpha \nu N(z, t) S dz dt$$

- Photon flux intensity $J(z, t) = N(z, t) \nu$

Transport equation of pulse traveling wave amplifier in a three-energy levels system

$$\eta_F = I, f_1 = f_2$$

$$\frac{\partial \Delta n(z, t)}{\partial t} = -2\sigma_{21} \Delta n(z, t) J(z, t) \quad (5.4.2)$$

$$\frac{\partial J(z, t)}{\partial t} + \nu \frac{\partial J(z, t)}{\partial z} = \nu \sigma_{21} \Delta n(z, t) J(z, t) - \alpha \nu J(z, t) \quad (5.4.3)$$

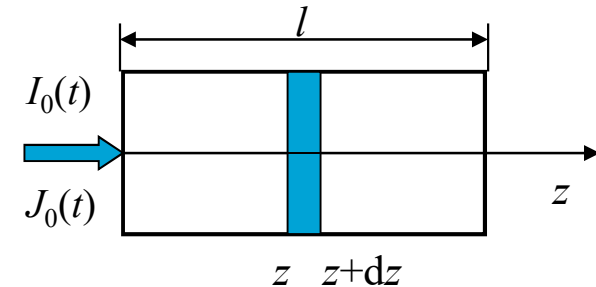
Boundary condition $J(0, t) = J_0(t)$; $\Delta n(z, t < 0) = \Delta n_0 (0 \leq z \leq l)$

Incident pulse waveform

Δn : spatial distribution (uniform)
before signal incidence

❖ Output energy and energy gain of pulse laser amplifier

Energy gain (G_E)



$$G_E \equiv \frac{E_l}{E_0} = \frac{J(l)}{J(0)} = \frac{\text{Total number of output photons per unit area}}{\text{Total number of input photons per unit area}}$$

Key: Get $J(l)$
from transport
equation

$$\frac{\partial \Delta n(z, t)}{\partial t} = -2\sigma_{21} \Delta n(z, t) J(z, t) \quad (5.4.2)$$

$$\frac{\partial J(z, t)}{\partial t} + v \frac{\partial J(z, t)}{\partial z} = v\sigma_{21} \Delta n(z, t) J(z, t) - \alpha v J(z, t) \quad (5.4.3)$$

Input signal energy $E_0 = h\nu S \int_0^{\tau'} J_0(t) dt = h\nu S J(0) \Leftrightarrow J(0) = \int_0^{\tau'} J_0(t) dt$

Output signal energy $E_l = h\nu S \int_0^{\tau'} J_0(l, t) dt = h\nu S J(l) \Rightarrow J(l) ?$

How to obtain the energy gain: No need to consider the behaviour of pulses at different times, mathematical treatment, time integration, Eq. 5.4.2 substitute 5.4.3 then integrate

$$\underbrace{\int_0^{\tau'} \frac{\partial J(z,t)}{\partial t} dt}_{=0} + \underbrace{v \int_0^{\tau'} \frac{\partial J(z,t)}{\partial z} dt}_{\Rightarrow \frac{dJ(z)}{dz}} = - \underbrace{\frac{v}{2} \int_0^{\tau'} \frac{\partial \Delta n(z,t)}{\partial t} dt}_{\Delta n_0 - \Delta n(z, \tau')} - \underbrace{\alpha v \int_0^{\tau'} J(z,t) dt}_{J(z)}$$

$$\frac{\partial \Delta n(z,t)}{\partial t} = -2\sigma_{21} \Delta n(z,t) J(z,t)$$

$$\frac{dJ(z)}{dz} = \frac{1}{2} [\Delta n_0 - \Delta n(z, \tau')] - \alpha J(z)$$

$$\int_{\Delta n_0}^{\Delta n(z, \tau')} \frac{d\Delta n(z,t)}{\Delta n(z,t)} = -2\sigma_{21} \int_0^{\tau'} J(z,t) dt$$

$$\Delta n(z, \tau') = \Delta n_0 e^{-2\sigma_{21} J(z)}$$

$$\frac{dJ(z)}{dz} = \frac{1}{2} [1 - e^{-2\sigma_{21} J(z)}] \Delta n_0 - \alpha J(z)$$

Nonlinear differential equation,
numerical solution

Some special cases for $J(z)$

$$\frac{dJ(z)}{dz} = \frac{1}{2} \left[1 - e^{-2\sigma_{21}J(z)} \right] \Delta n_0 - \alpha J(z)$$

(1) Weak incident signal $\sigma_{21}J(z) \ll 1$

Expand and only consider
the first two items

$$1 - 2\sigma_{21}J(z)$$

$$\frac{dJ(z)}{dz} \approx (\sigma_{21}\Delta n_0 - \alpha)J(z) \Rightarrow J(l) = J(0)e^{(\sigma_{21}\Delta n_0 - \alpha)l}$$

Weak signal energy gain

$$G_E = J(l)/J(0) = \exp[(\sigma_{21}\Delta n_0 - \alpha)l]$$

Conclusion:

- G_E independent of signal intensity,
- $l \uparrow$ or $\Delta n_0 \uparrow \rightarrow G_E \uparrow$; G_E **exponentially** increase with l and Δn_0

(2) Intensive incident signal $\sigma_{21}J(z) \gg 1 \Rightarrow e^{-\sigma_{21}J(z)} \sim 0$

$$\frac{dJ(z)}{dz} = \frac{1}{2} \left[1 - e^{-2\sigma_{21}J(z)} \right] \Delta n_0 - \alpha J(z) \approx \frac{1}{2} \Delta n_0 - \alpha J(z)$$

can be re-written as

$$\frac{dJ(z)}{J(z) - (\Delta n_0/2\alpha)} = -\alpha dz$$

The total number of photons per unit area can be obtained after integration $J(l)$

$$J(l) = \frac{\Delta n_0}{2\alpha} + J(0)e^{-\alpha l} - \frac{\Delta n_0}{2\alpha}e^{-\alpha l} = J(0)e^{-\alpha l} + \frac{\Delta n_0}{2\alpha}(1 - e^{-\alpha l})$$

$$\alpha \ll 1 \xrightarrow{l \text{ short}} \alpha l \ll 1$$

$$(1 - \alpha l)$$

intensive signal
energy gain

$$J(l) = J(0) + \frac{\Delta n_0}{2\alpha}(1 - e^{-\alpha l}) \approx J(0) + \frac{\Delta n_0 l}{2} \Rightarrow G_E = 1 + \frac{\Delta n_0 l}{2J(0)}$$

Conclusion:

- $J(0) \uparrow \rightarrow G_E \downarrow$ Gain saturation
- Intensive signal incidence, G_E **linearly** increase with l and Δn_0 , NOT exponentially increase, because Δn_0 will deplete significantly.

(3) l long enough, the intensity of the incident light signal is within the two

$$J(l) = J(0)e^{-\alpha l} + \frac{\Delta n_0}{2\alpha}(1 - e^{-\alpha l})$$

$$G_E = \frac{\Delta n_0}{2\alpha J(0)}(1 - e^{-\alpha l}) + e^{-\alpha l}$$

At the beginning, it is similar with the weak signal case, that is, exponentially increase with l . With the increase of the transmission distance l , once the pulse energy is strong enough, gain saturation appears. When the output pulse energy approaches saturation, it does not increase anymore.

G_E approach stable

$$G_E = \frac{\Delta n_0}{2\alpha} \frac{1}{J(0)}$$

At this moment, the energy gain is **independent** of the length of the amplifier. The maximum photon number density of amplifier output is independent of the input energy, only related to Δn_0 and α .

(4) Lossless laser amplifier ($\alpha=0$)

$$\frac{dJ(l)}{dl} = \frac{1}{2} \left[1 - e^{-2\sigma_{21}J(l)} \right] \Delta n_0 - \alpha J(l)$$

Get $J(l), G_E$ by integrating the above equation

Lossless laser amplifier

$$J(l) = \frac{1}{2\sigma_{21}} \ln \left[1 + \left(e^{2\sigma_{21}J(0)} - 1 \right) e^{\sigma_{21}\Delta n_0 l} \right]$$

$$G_E = \frac{1}{2\sigma_{21}J(0)} \ln \left[1 + \left(e^{2\sigma_{21}J(0)} - 1 \right) e^{\sigma_{21}\Delta n_0 l} \right]$$

Output energy and energy gain increase with l regardless of the intensity of the incident signal.

Summary: for pulse laser amplifier with α

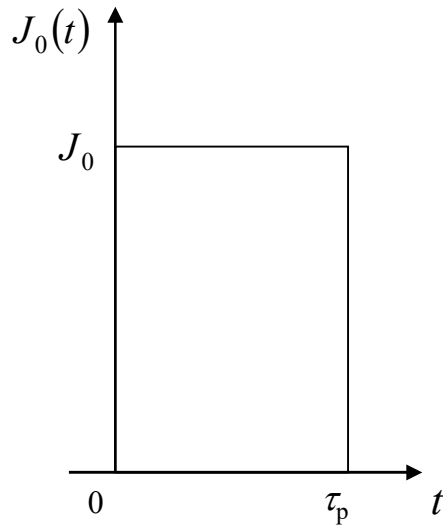
$\left\{ \begin{array}{l} \text{Weak signal} \\ \text{Intensive signal} \\ l \text{ long enough} \end{array} \right.$	$G_E = e^{(\sigma_{21}\Delta n_0 - \alpha_i)l}$	Exponentially
	$G_E = 1 + \frac{\Delta n_0 l}{2J(0)}$	Linearly
	$G_E = \frac{\Delta n_0}{2\alpha_i} \frac{1}{J(0)}$	When saturated, α has great impact on the output energy

The most effective way to improve the output energy of pulse amplifier:

$\Delta n_0 \uparrow, \alpha \downarrow$
rather than simply increase l

❖ Power gain (G_p) and pulse width narrowing

Power gain: to analyze the output pulse power and waveform, we need the unsteady solution $J(z,t)$



$$G_p(t) = \frac{1}{1 - \left(1 - e^{-\sigma_{21}\Delta n_0 l}\right) e^{-2\sigma_{21}J_0 t}}$$

$$t = 0 \quad G_p(0) = \exp(\sigma_{21}\Delta n_0 l)$$

$$t \neq 0 \quad 2\sigma_{21}J_0 t \ll 1 \Rightarrow G_p(t) \approx \exp(\sigma_{21}\Delta n_0 l)$$

Weak signal: linearly amplify, no deformation

Intensive signal: Gain saturation, $t \uparrow \rightarrow G_p(t) \downarrow$
nonlinearly amplify, pulse deforms and becomes narrower

Physical reason for the distortion of the input strong pulse

- The front and top of the rectangular optical pulse have different gains: the front has exponential power gain while the top gain decreases with the increase of t .

